Mahindra University Hyderabad
École Centrale School of Engineering
Minor II Regular Examination
(Batch 2023)

Program: B. Tech **Branch: All (except, CB & BT)** **Year: I** **Semester: I**
Subject: Chemistry - I (CH 1101)

Date: 09/11/2023 **Start Time: 10.00 AM**
Time Duration: 1.5 Hours **Max. Marks: 30**

Instructions

1. Answer all questions.
2. No clarifications will be entertained during the examination.
3. If any information is missing, make appropriate assumptions and proceed.

Table 1: Useful physical constants and conversion factors

Speed of light (c)	=	2.997×10^{8} m s^{-1}
Mass of electron (m_e)	=	9.109×10^{-31} kg
Planck constant (h)	=	6.626×10^{-34} J s
Boltzmann constant (k_B)	=	1.381×10^{-23} J K^{-1}
Avogadro number (N_A)	=	6.023×10^{23} mol^{-1}
Elementary charge (e)	=	1.602×10^{-19} C
Mass of proton (m_p)	=	1.672×10^{-27} kg
Rydberg Constant (R_H)	=	109677 cm^{-1}
1 eV	=	1.602×10^{-19} J
1 E_h	=	27.211 eV

Q1. (Marks: [6])

For a particle in a state having the following wavefunction:

$$\psi = N \sin \frac{\pi x}{a}$$

in the range $x = 0$ to $x = a$. Normalize the wavefunction and find the normalization constant N.

Q2. (Marks: [3+3=6])

For a proton confined in a 1-dimensional box of length 1.00 nm, find the energy of the 2nd quantum state (n=2). What should be the length of the box if the proton is replaced by an electron to have the same energy?

Q3. (Marks: [6])

Assume the π-electrons of benzene freely moving in a circular ring of carbon atoms. Calculate the diameter of this "ring" if electronic transition occurs from the ground-state (highest-occupied) to the next excited-state (lowest-unoccupied) at the wavelength of 212 nm.

Q4. (Marks: [2+4=6])

(i) State whether each of the homonuclear diatomic molecular orbitals (MOs) given below is bonding or antibonding.

$$\text{(a) } \sigma_u, \quad \text{(b) } \pi_u$$

Draw the MO for each case.

(ii) Find the bond order of HF molecule using molecular orbital energy level diagram.

Q5. (Marks: [6])

UV-Visible spectrum of a solution of unknown concentration of an organic compound (molecular weight = 200 g/mol) was measured using a 1.0 cm long sample cell. The spectrum shows that at a wavelength of 350 nm, only 39.8% of light is transmitted by this solution. Determine the amount (in mg) of this organic molecule present in 1.0 L of this solution. The molar absorption coefficient (ϵ) of this molecule at 350 nm is 800 L mol^{-1} cm^{-1}.